

$$1) P(Y > y) = e^{-y/\mu}$$

$$a) P(Y > 4/0) = e^{-4/2} = 0.1353$$

$$b) P(Y > 4/1) = e^{-4/5} = 0.4493$$

$$c) P(Y > 4) = P(Y > 4/0)P(0) + P(Y > 4/1)P(1) = 0.2295$$

$$d) P(1/Y > 4) = \frac{P(Y > 4/1)P(1)}{P(Y > 4)} = 0.587$$

$$e) P(0/Y \leq 4) = \frac{f(4/0)P(0)}{f(4/0)P(0) + f(4/1)P(1)} \Rightarrow 0.637$$

$$2) a) Z = X_1^2 + X_2^2 \quad f_{X_1, X_2}(x_1, x_2) = \frac{1}{2\pi} e^{\left(-\frac{x_1^2 + x_2^2}{2}\right)}$$

$$r^2 = x_1^2 + x_2^2$$

$$x_1 = r \cos \theta$$

$$x_2 = r \sin \theta$$

$$|J| = r$$

Joint PDF

$$f_{R, \theta}(r, \theta) = f_{X_1, X_2}(r \cos \theta, r \sin \theta) \cdot r = \frac{1}{2\pi} e^{-r^2/2} r$$

margi $f_R(r) = \int_{-\pi}^{\pi} f_{R, \theta}(r, \theta) d\theta = r e^{-r^2/2} \quad r \geq 0$

$$Z = r^2$$

$$r = \sqrt{z}$$

$$\frac{dr}{dz} = \frac{1}{2\sqrt{z}}$$

$$f_z(z) = f_R(\sqrt{z}) \left| \frac{dr}{dz} \right| = \left(\sqrt{z} e^{-z/2} \right) \frac{1}{2\sqrt{z}}$$

$$f_z(z) = \frac{1}{2} e^{-z/2} \quad \lambda = \frac{1}{2}$$

$$E[Z] = \frac{1}{\lambda} = 2$$

b) Z is indep. of θ

$$f_{R,\theta}(r,\theta) = \underbrace{r e^{-r^2/2}}_{f_R(r)} \cdot \underbrace{\frac{1}{2\pi}}_{f_\theta(\theta)} = f_R(r) f_\theta(\theta).$$

$$3) f_{Y|X}(y|0) = \begin{cases} \frac{1}{2} & 0 \leq y \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

$$f_{Y|X}(y|1) = \begin{cases} \frac{1}{2} & 1 \leq y \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

$$b) f_Y(y) = f_{Y|X}(y|0) P(X=0) + f_{Y|X}(y|1) P(X=1)$$

$$0 \leq y \leq 1 \quad f_Y(y) = 0.6 \times \frac{1}{2} = 0.3$$

$$1 \leq y \leq 2 \quad f_Y(y) = 0.6 \times \frac{1}{2} + 0.4 \times \frac{1}{2} = 0.3 + 0.2 = 0.5$$

$$2 \leq y \leq 3 \quad f_Y(y) = 0.4 \times \frac{1}{2} = 0.2$$

$$f_Y(y) = \begin{cases} 0.3 & 0 \leq y < 1 \\ 0.5 & 1 \leq y \leq 2 \\ 0.2 & 2 \leq y \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

$$c) P(1 < Y < 2) = \int_1^2 0.5 dy = 0.5$$

$$d) P(X=1 | Y=1.5) = \frac{0.5 \times 0.4}{0.5} = 0.4$$

$$4) a) E[X(t)] = A \int_{-\pi}^{\pi} \cos(\omega t + \theta) \frac{1}{2\pi} d\theta = 0$$

$$b) X(t_1) X(t_2) = A^2 \cos(A) \cos(B) \quad R_X = \frac{A^2}{2} \cos(\omega(t_1 - t_2))$$

$$c) \text{Mean} = \text{const.}$$

$$R_X \text{ depends only on } t_1 - t_2 \Rightarrow \text{WSS}$$

$$5) \text{PSD after filter} = S_n(f) = \frac{N_0}{2} \text{ for } 2\omega$$

$$\text{Tot. Power} = P = 2 \times \left(\frac{N_0}{2} \right) \times 2\omega$$

$$n_c(t) = n(t) \cos 2\pi f_c t + \hat{n}(t) \sin 2\pi f_c t$$

$$n_s(t) = \hat{n}(t) \cos 2\pi f_c t - n(t) \sin 2\pi f_c t$$

$$E n_c(\tau) = E [n_c(t + \tau) n_c(t)]$$

$$R_{n_c}(\tau) = R_n(\tau) \cos 2\pi f_c \tau + \hat{R}_n(\tau) \sin 2\pi f_c \tau$$

$$S_{nc}(f) = S_{ns}(f)$$

$$P_{nc} = P_{ns} = \int_{-\infty}^{\infty} S_{nc}(f) df = 2 N_0 W.$$

$$b) R_{nc, ns}(\tau) = E[n_c(t+\tau) n_s(t)].$$

$$= R_n(\tau) \sin \tau - \hat{R}_n \cos \tau.$$

$$S_{nc, ns}(f) = \mathcal{F}[R_{nc, ns}(\tau)] = \mathcal{F}[R_n(\tau) \sin \tau - \hat{R}_n \cos \tau].$$

$$V(t) \triangleq \sqrt{n_c^2(t) + n_s^2(t)} \rightarrow \text{Rayleigh's distribution.}$$

$$f(V, \sigma) = \frac{V}{\sigma^2} \cdot e^{-V^2/2\sigma^2}$$

$$\sigma^2 = P_{nc} = 2 N_0 W$$

$$6. (a) E[U(t)] = A_c e^{-R\phi(t)/2} \cos(\omega_c t) \rightarrow \text{not stable}$$

$$(b) \bar{R}_u(\tau) = \frac{A_c^2}{2} \cos(\omega_c \tau) e^{-(R\phi(0) - R\phi(\tau))}$$

$$(c) \text{F.T. } S_u(f) = \frac{A_c^2}{4} (G(f-f_c) + G(f+f_c))$$

7) from previous.

$$P = \frac{1}{T} \int |Y_T(f)|^2 df \quad \Rightarrow \quad S_{yy}(f) = \lim_{T \rightarrow \infty} \frac{1}{T} |Y_T(f)|^2$$

$$b) \frac{1}{b} |Y_{T_0}(f)|^2 = \frac{1}{T T_0} \{ y_{T_0}^*(-t_0) + y_{T_0}(t_0) \}.$$

$$\tilde{R}_{yy}(\tau) = \lim_{T_0 \rightarrow \infty} \frac{1}{T T_0} \int y_{T_0}^*(t-\tau) y_{T_0}(t) dt$$

$$S_{yy}(f) = \lim_{T_0 \rightarrow \infty} \frac{1}{T} |Y_0(f)|^2 = \int_{-\infty}^{\infty} \tilde{R}_{yy}(\tau) e^{-j2\pi f\tau} d\tau$$

$$8. a) E[y(t)] = \sum_k E[a_k] g(t - kT)$$

$$E[a_k] = 0 \Rightarrow E[y(t)] = 0.$$

mean const. $\Rightarrow$ periodic with any period.

$$b) R_y = \sum_k \sum_m E[a_k \cdot a_m^*] g(t - kT)$$

$$E[a_k a_m^*] = \begin{cases} \sigma_a^2 & k=m \\ 0 & k \neq m. \end{cases}$$

$$R_y(t, \tau) = \sigma_a^2 \sum_k g(t - kT) g^*(t - \tau - kT)$$

$$c) R_y(t + T, \tau) = \sigma_a^2 \sum_k g(t + T - kT) g^*(t + T - \tau - kT)$$

$$n = k - 1. \quad R_y(t + T, \tau) = R_y(t, \tau).$$

Cyclostationary with period T .

$$9) a) S_n(f) = \frac{N_0}{2} \quad Z(f) = (h * h^*)(f)$$

$$Z[n] = Z(nT_s).$$

$$\text{cov}(Z[n], Z[m]) = E[Z(nT_s) Z^*(mT_s)] = R_z((n-m)T_s)$$

$$\text{low-pass } h_{LP}(f) = h^*(-f)$$

$$\text{cov}(Z[n], Z[m]) = \frac{N_0}{2} \int h(t) h^*(t - (n-m)T_s) dt$$

$$4) \text{cov}(Z[n], Z[m]) = \frac{N_0}{2} (h * h_{m.f}) ((n-m)T_s)$$

$$Z(f) = (h * h_{m.f})(f)$$

$$Q(f) = H(f) H_{m.f}(f) = H(f) H^*(f) = |H(f)|^2$$

criterion $\Rightarrow H(f)$ square root nyquist filter

10) a) triangle f^n convolution.

$$F_1(f) = \sin^2(\pi f) \rightarrow \text{sym \& non negative}$$

$$d) F_3(f) = F_1(f) - \frac{1}{2} F_1(f) / [e^{-j2\pi f} + e^{j2\pi f}] = \sin^2(\pi f) (1 - \cos 2\pi f)$$

$\hookrightarrow$ sym. non negative & hence $F_3(f)$ is a valid PSD.

$$11. S_X(f) = \int_{-\infty}^{\infty} e^{-a\tau} e^{-j2\pi f\tau} d\tau + \int_{-\infty}^{\infty} e^{a\tau} e^{-j2\pi f\tau} d\tau$$

$$= \frac{e^{-(a+j2\pi f)\tau}}{-(a+j2\pi f)} \Big|_{-\infty}^{\infty} + \frac{e^{+(a-j2\pi f)\tau}}{a-j2\pi f} \Big|_{-\infty}^{\infty}$$

$$= \frac{2a}{a^2 + 4\pi^2 f^2}$$

output power $\int_{-\infty}^{\infty} S_X(f) df$

$$= \int_{-\tan^{-1}(\frac{2\pi f}{a})}^{\tan^{-1}(\frac{2\pi f}{a})} \frac{2a}{a^2 + 4\pi^2 f^2} df$$

$$= \frac{2}{\pi} \tan^{-1}\left(\frac{2\pi f}{a}\right) \Big|_{-\tan^{-1}(\frac{2\pi f}{a})}^{\tan^{-1}(\frac{2\pi f}{a})}$$

$$12. R_z(t + \tau) \approx E[z(t + \tau) z(t)]$$

$$\Rightarrow E[(X(t + \tau) + Y(t + \tau))(X(t) + Y(t))]$$

$$= R_x(\tau) + R_y(\tau) + R_{xy}(\tau) + R_{xy}(-\tau)$$

$$13. m_y = m_x \int_{-\infty}^{\infty} \frac{1}{\tau} d\tau \Rightarrow$$

$$R_y(\tau) = R_{xy}(\tau) * \frac{1}{\tau} = -R_x(\tau) = R_x(-\tau)$$

$$\hookrightarrow \hat{x}(t) = -x(t)$$

14. No, the converse is not necessarily true,

$$y(t) = x(t) * h(t)$$

$$x(t) = m(t) + w(t)$$

↓
determine stationarity from

filter time-variant s;

$$\hookrightarrow y(t) = w(t) * h(t)$$

↓
stationary

Since $w(t) \rightarrow$ stationary $y(t)$
is stationary even when
 $x(t)$ is not stationary.

15. a) $x_1(t) = (n * h)(t)$

where $H(f) = j2\pi f \Rightarrow \frac{d}{dt}$

$$S_{x_1}(f) = 4\pi^2 f^2 I_{(-1,1)}(f)$$

b) $x_2(t) = (n * g)(t)$ $g(t) = \frac{1}{d} (\delta(t) - \delta(t-d))$

$$G(f) = \frac{1 - e^{-j2\pi f d}}{d}, \text{ now } |G(f)|^2 = \frac{1 - e^{-j2\pi f d}}{d} \cdot \frac{1 - e^{j2\pi f d}}{d}$$

$$|G(f)|^2 = \frac{4 \sin^2(\pi f d)}{d^2}$$

$$I = \frac{1}{2} \quad S_{x_2}(f) = 4\pi^2 f^2 \sin^2\left(\frac{f}{2}\right) I_{(-1,1)}(f)$$

i) power for $R_1(0) = \int_{-1}^1 4\pi^2 f^2 df = \frac{8\pi^2}{3} \approx 26.3$

for $x_2 (d = \frac{1}{2})$ $R_{x_2}(0) = \int_{-1}^1 4\pi^2 f^2 \sin^2\left(\frac{f}{2}\right) df \approx 16.0$

6. (a) $h(t) = I_{(-1,1)}(t)$

$$H(f) = 2 \operatorname{sinc}(2f)$$

$$S_y(f) = S_n(f) |H(f)|^2$$

$$S_y(f) = 4 \sin^2(2f)$$

(b) $\operatorname{cov}(y(t_1), y(t_2)) = R_y(t_1 - t_2)$
 $= r_y(t_1 - t_2)$

$$R_y(\tau) = \sigma^2 (h * h_{\text{rev}})(\tau)$$

$$Y = [y(1), y(2), y(3)]^T$$

$$C = \begin{pmatrix} R_y(0) & R_y(1) & R_y(2) \\ \vdots & \ddots & \vdots \end{pmatrix} = \begin{pmatrix} 2 & 1 & 0 \\ 2 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix} \sim N(0, C)$$

$$1) Z = y(1) - 2y(2) + y(3)$$

$$Z = a^T y$$

$$a = (1, -2, 1)^T$$

y -gaussian

$$Z \sim N(0, \sigma_z^2)$$

$$\sigma_z^2 = a^T R a$$

$$P(Z > 10) = Q\left(\frac{10}{\frac{\sigma_z}{2}}\right) = Q(15)$$